Exploratory Data Analysis

Website Quality & Conversion Gap Scorer

Course	Machine Learning AIN-373 — Spring 2026
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Dataset	data\labeled\dataset.csv
Samples	221 websites
Features	41 across 4 dimensions (UX, Content, Technical, Trust)
Target	score_total $\in$ [0, 100], μ=43.99, σ=10.97
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1. Dataset Overview

Total rows	221
Features	41 (continuous: 27, boolean: 14)
UX dimension	10 features
Content dimension	10 features
Technical dimension	11 features
Trust dimension	10 features
Labels	score_ux, score_content, score_technical, score_trust (0-25 each) + score_tota
NaN cells in feature matrix	612 / 9061 (6.75%)

2. Target Distribution & Tier Breakdown

Mean	43.99
Median	44.46
Std	10.97
Min / Max	16.60 / 71.01
Skewness	-0.195
Kurtosis	-0.601

Tier counts

Tier	Count	Share
Poor (0-25)	11	5.0%
Average (26-50)	138	62.4%
Good (51-75)	72	32.6%
Excellent (76-100)	0	0.0%

Sub-score statistics

Sub-score	Mean	Std	Min	Max
score_ux	9.17	6.13	0.00	20.00
score_content	9.80	5.07	0.00	22.40
score_technical	14.06	2.70	8.75	24.65
score_trust	7.89	5.60	0.00	24.25

3. Missing-Value Analysis

5 of 41 columns have missing values. PageSpeed columns dominate (crawled without API key).

Feature	Dimension	% Missing
tech_pagespeed_desktop	technical	56.1%
tech_lcp_seconds	technical	55.2%
tech_pagespeed_mobile	technical	55.2%
tech_cls	technical	55.2%
tech_tbt_seconds	technical	55.2%

4. Top Predictive Features

Features sorted by absolute Pearson correlation with **score_total**. These are the strongest linear signals; tree models like XGBoost can also exploit nonlinear interactions invisible here.

#	Feature	Dim	Pearson r	Spearman r
1	content_heading_hierarchy_score	content	+0.617	+0.585
2	content_h1_unique	content	+0.570	+0.578
3	trust_has_pricing	trust	+0.541	+0.542
4	trust_testimonials	trust	+0.528	+0.528
5	ux_cta_total	ux	+0.509	+0.601
6	content_value_prop_detected	content	+0.506	+0.511
7	content_paragraph_length_std	content	+0.484	+0.577
8	tech_pagespeed_mobile	technical	+0.468	+0.458
9	trust_client_logos	trust	+0.414	+0.400
10	ux_has_hero_section	ux	+0.391	+0.396
11	content_word_count	content	+0.352	+0.523
12	tech_tbt_seconds	technical	-0.272	-0.238
13	ux_cta_above_fold	ux	+0.256	+0.318
14	content_h1_count	content	+0.255	+0.483
15	trust_has_video	trust	+0.252	+0.259
16	trust_privacy_link	trust	+0.233	+0.234
17	tech_lcp_seconds	technical	-0.189	-0.319
18	tech_cls	technical	-0.176	-0.041
19	tech_schema_markup	technical	+0.167	+0.171
20	trust_email_visible	trust	+0.159	+0.161

5. Multicollinearity & Constant Features

Feature pairs with |r| ≥ 0.85

Feature A	Feature B	Dim A	Dim B	Pearson r
content_flesch_reading_ease	content_flesch_kincaid_grade	content	content	-1.000

Constant / near-constant features

These features have ≥95% of values concentrated in a single bucket (or near-zero variance). Drop before training.

Feature	Dimension
ux_viewport_meta	ux
ux_responsive_breakpoints	ux
tech_https	technical

Feature	Dimension
tech_external_js_count	technical
trust_address_visible	trust

6. Skewness & Outliers

Features with |skew| > 1 (log1p transform candidates)

Feature	Dim	Skew
content_flesch_reading_ease	content	-14.86
content_flesch_kincaid_grade	content	+14.86
ux_responsive_breakpoints	ux	+10.44
tech_external_css_count	technical	+7.70
trust_social_proof_count	trust	+7.50
ux_image_text_ratio	ux	+4.82
ux_form_field_count	ux	+4.71
content_avg_sentence_length	content	+3.96
tech_cls	technical	+3.82
tech_lcp_seconds	technical	+3.68
content_h1_count	content	+3.68
tech_page_size_kb	technical	-3.57
content_meta_description_length	content	+3.45
trust_social_link_count	trust	+3.45
ux_whitespace_ratio	ux	+3.29
ux_cta_above_fold	ux	+2.51
content_word_count	content	+2.25
ux_nav_link_count	ux	+2.05
tech_tbt_seconds	technical	+2.01
ux_total_links	ux	+1.94
ux_cta_total	ux	+1.72
content_paragraph_length_std	content	+1.48

Top 15 features by IQR outlier count

Feature	Dim	Outlier count
content_meta_description_length	content	39
content_flesch_kincaid_grade	content	31
content_flesch_reading_ease	content	27
content_avg_sentence_length	content	23
ux_whitespace_ratio	ux	22
ux_form_field_count	ux	21
ux_cta_above_fold	ux	19
ux_nav_link_count	ux	18
tech_cls	technical	17

Feature	Dim	Outlier count
ux_image_text_ratio	ux	16
ux_total_links	ux	9
content_paragraph_length_std	content	9
content_h1_count	content	8
tech_tbt_seconds	technical	7
ux_cta_total	ux	6

7. Sub-Score Inter-Correlation

Off-diagonal Pearson r between the four dimension labels. Low values indicate the four sub-scores carry independent information — desirable for decomposing the composite score.

	score_ux	score_content	score_technical	score_trust
score_ux	+1.00	+0.27	-0.10	+0.24
score_content	+0.27	+1.00	+0.13	+0.45
score_technical	-0.10	+0.13	+1.00	+0.07
score_trust	+0.24	+0.45	+0.07	+1.00

8. Preprocessing Recommendations

- **Missing values.** PageSpeed columns are >30% missing (tech_pagespeed_mobile (55%), tech_pagespeed_desktop (56%), tech_lcp_seconds (55%), tech_cls (55%), tech_tbt_seconds (55%)). **Action:** keep as NaN — XGBoost handles missing natively. If switching to a non-tree model, apply median imputation per column.
- **Skew correction.** 22 continuous features have |skew| > 1. **Action:** apply `np.log1p` before training (particularly: `content_flesch_reading_ease`, `content_flesch_kincaid_grade`, `ux_responsive_breakpoints`, `tech_external_css_count`, `trust_social_proof_count`...).
- **Multicollinearity.** within-dimension: `content_flesch_reading_ease` ↔ `content_flesch_kincaid_grade` (r=-1.00) → drop one.
- **Constant / near-constant features.** Drop before training: `ux_viewport_meta`, `ux_responsive_breakpoints`, `tech_https`, `tech_external_js_count`, `trust_address_visible`.
- **Class imbalance.** Poor (0-25), Excellent (76-100) tiers hold <10% of samples. **Action:** stratify train/test split on tier label and use stratified k-fold CV.
- **Scaling.** Continuous features → StandardScaler; booleans → passthrough. XGBoost is scale-invariant, so this is optional for the deployed model but required if benchmarking against linear models or k-NN.
- **Train/test protocol.** 80/20 split with `stratify=tier`; 5-fold stratified CV for hyperparameter tuning. Hold out final 20% only for end-of-project metrics.

9. Figure Appendix

All 14 figures generated by `scripts/eda.py`. Source PNGs live in `data/eda/figures/`.

01 target total distribution

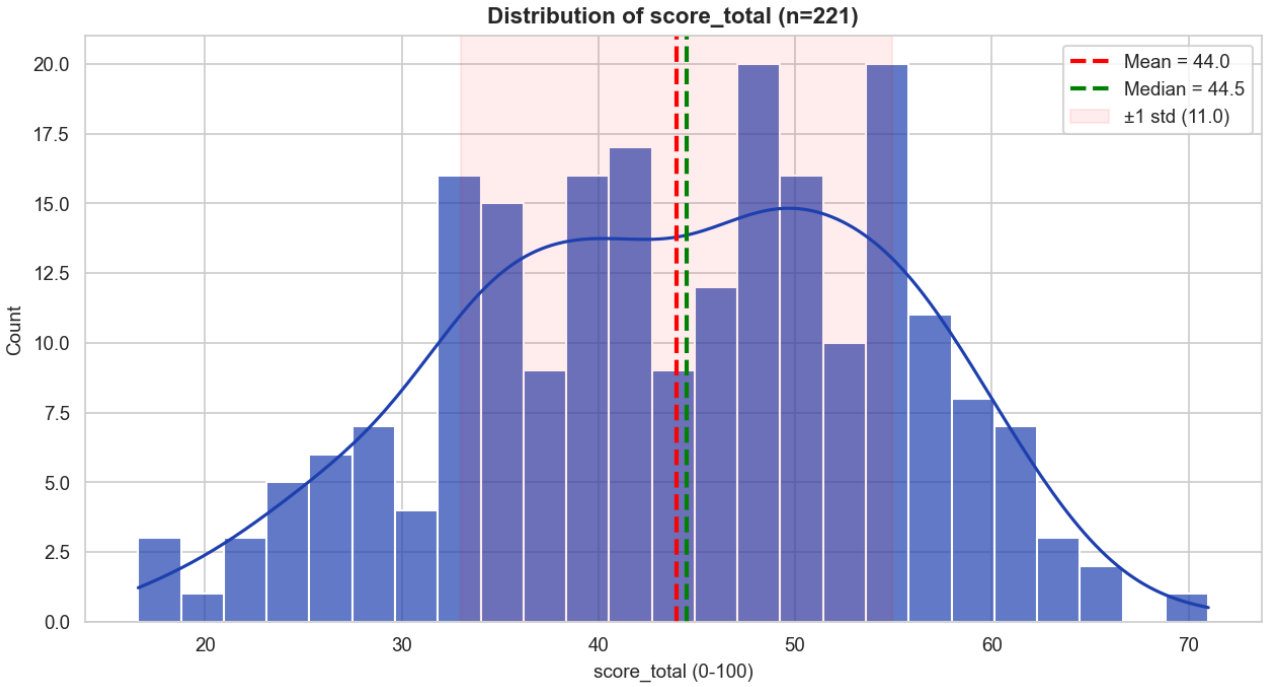

Figure 1 — Distribution of score_total. The narrow [16.6, 71.0] range and slight left-skew (skew=–0.20) indicate no website in the dataset reached the Excellent tier.

02 subscore distributions

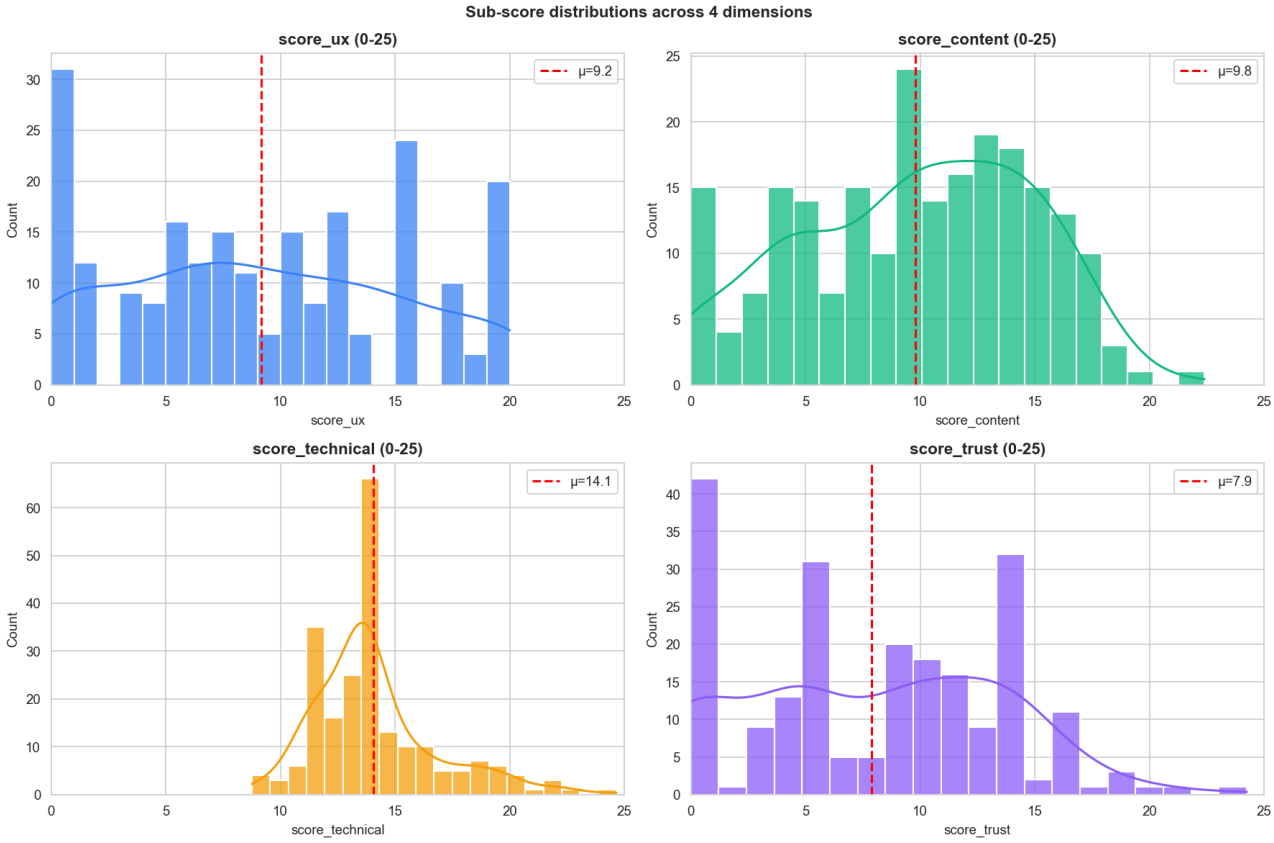

Figure 2 — Sub-score distributions per dimension. Technical and Trust sub-scores cluster lower (median ≈ 10–14) while UX and Content span more of the 0–25 range.

03 tier pie

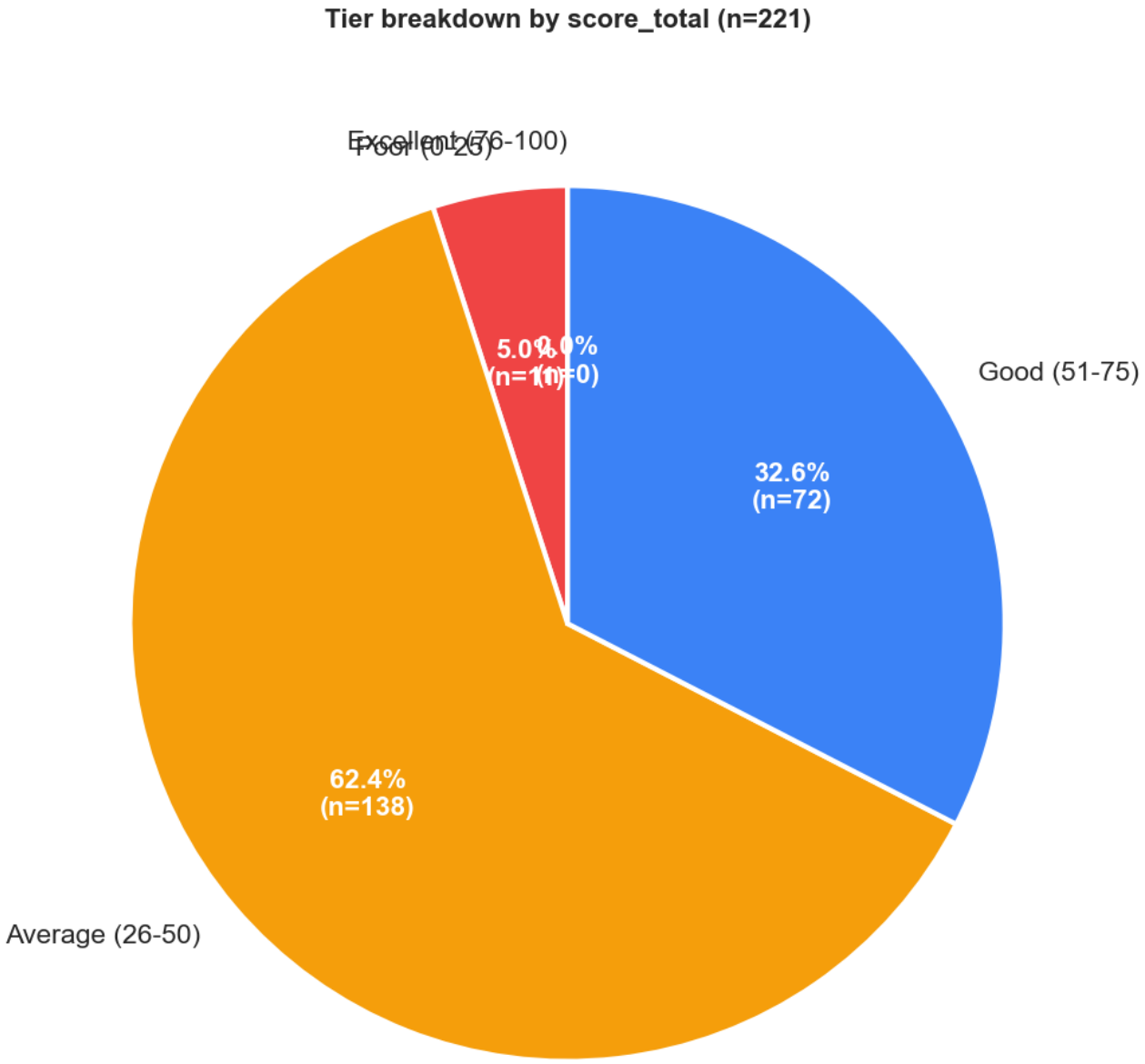

Figure 3 — Tier breakdown. Average (62.4%) dominates, Good (32.6%) is solid, but Poor (5.0%) is sparse and Excellent (0%) is absent — class imbalance to address via stratified splitting.

04 missing values

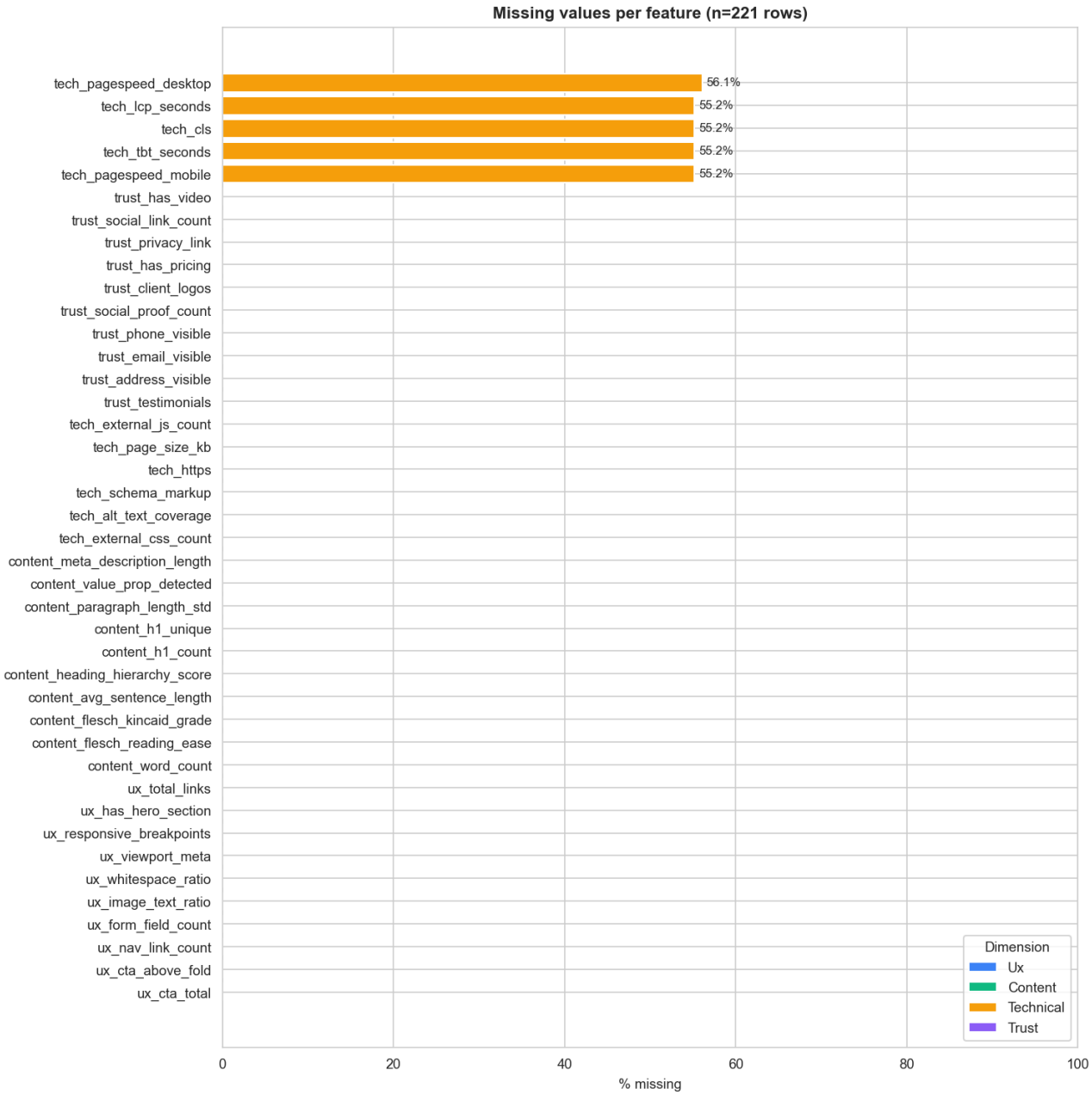

Figure 4 — % missing per feature. Five PageSpeed columns dominate (~55% each) due to crawling without a Google API key; all other features are complete.

05a ux feature distributions

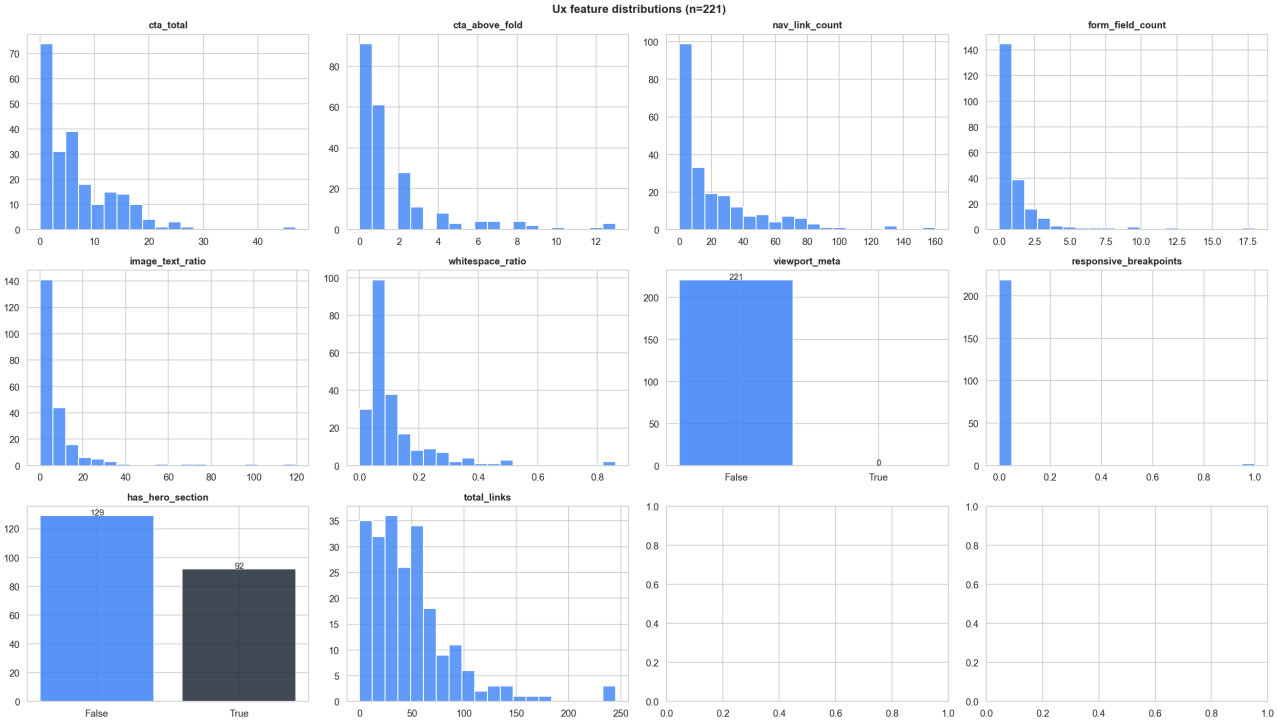

Figure 5a — UX feature distributions. Most CTA/link counts are zero-heavy; viewport_meta and responsive_breakpoints are degenerate.

05b content feature distributions

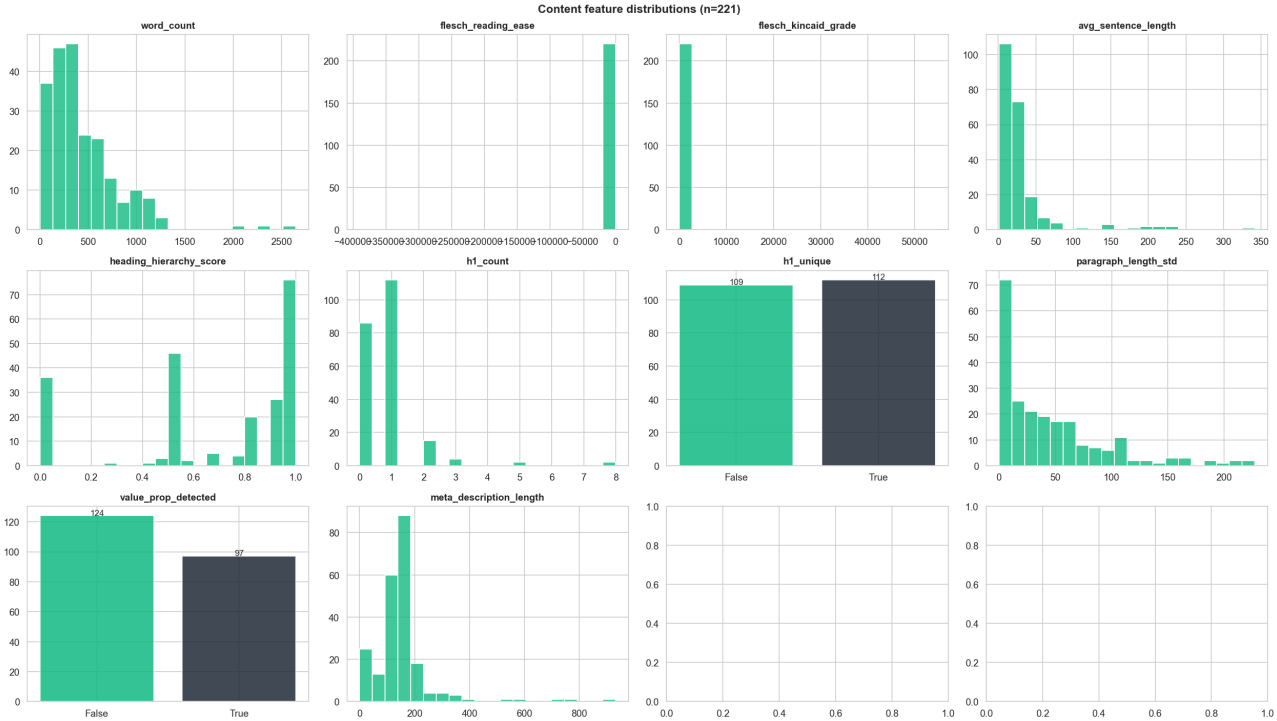

Figure 5b — Content features. Word counts and readability are roughly bell-shaped; h1_count and h1_unique skew toward zero.

05c technical feature distributions

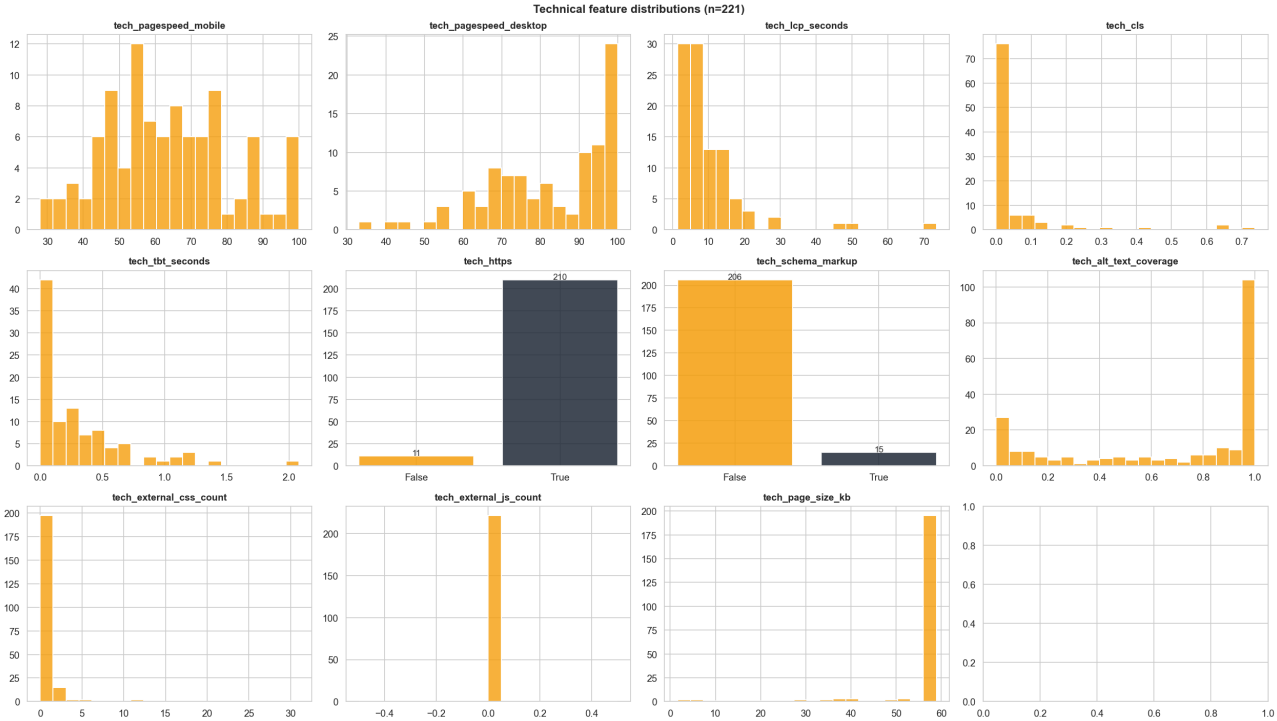

Figure 5c — Technical features. PageSpeed and Core Web Vitals are sparse (NaN-heavy). HTTPS and external_js_count are near-constant.

05d trust feature distributions

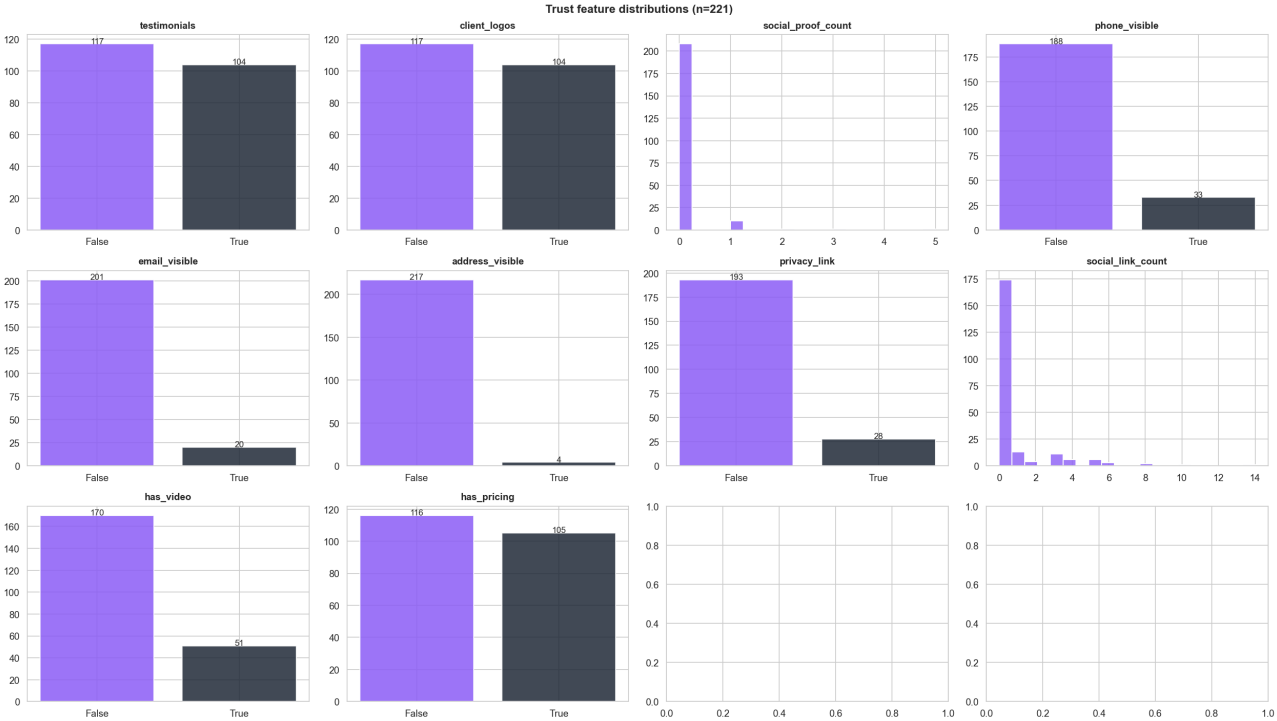

Figure 5d — Trust features. Most booleans skew False; address_visible is near-constant (almost always missing/false).

06 boolean feature counts

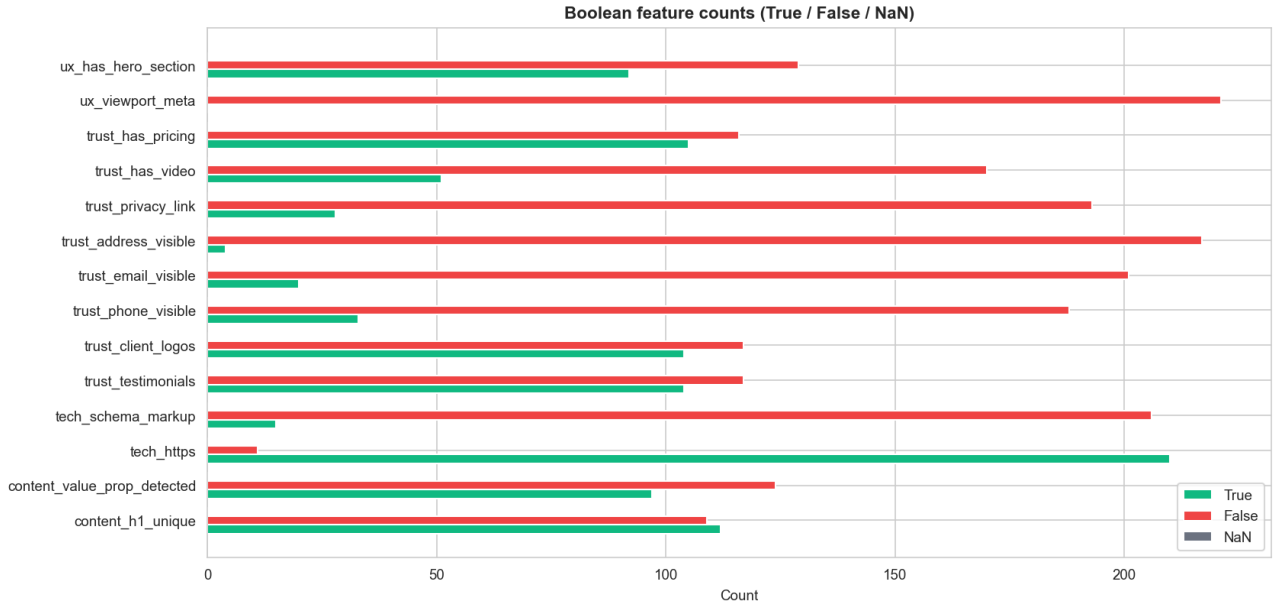

Figure 6 — True / False / NaN counts for every boolean feature. Reveals which boolean signals carry usable variance vs. degenerate.

07 correlation heatmap

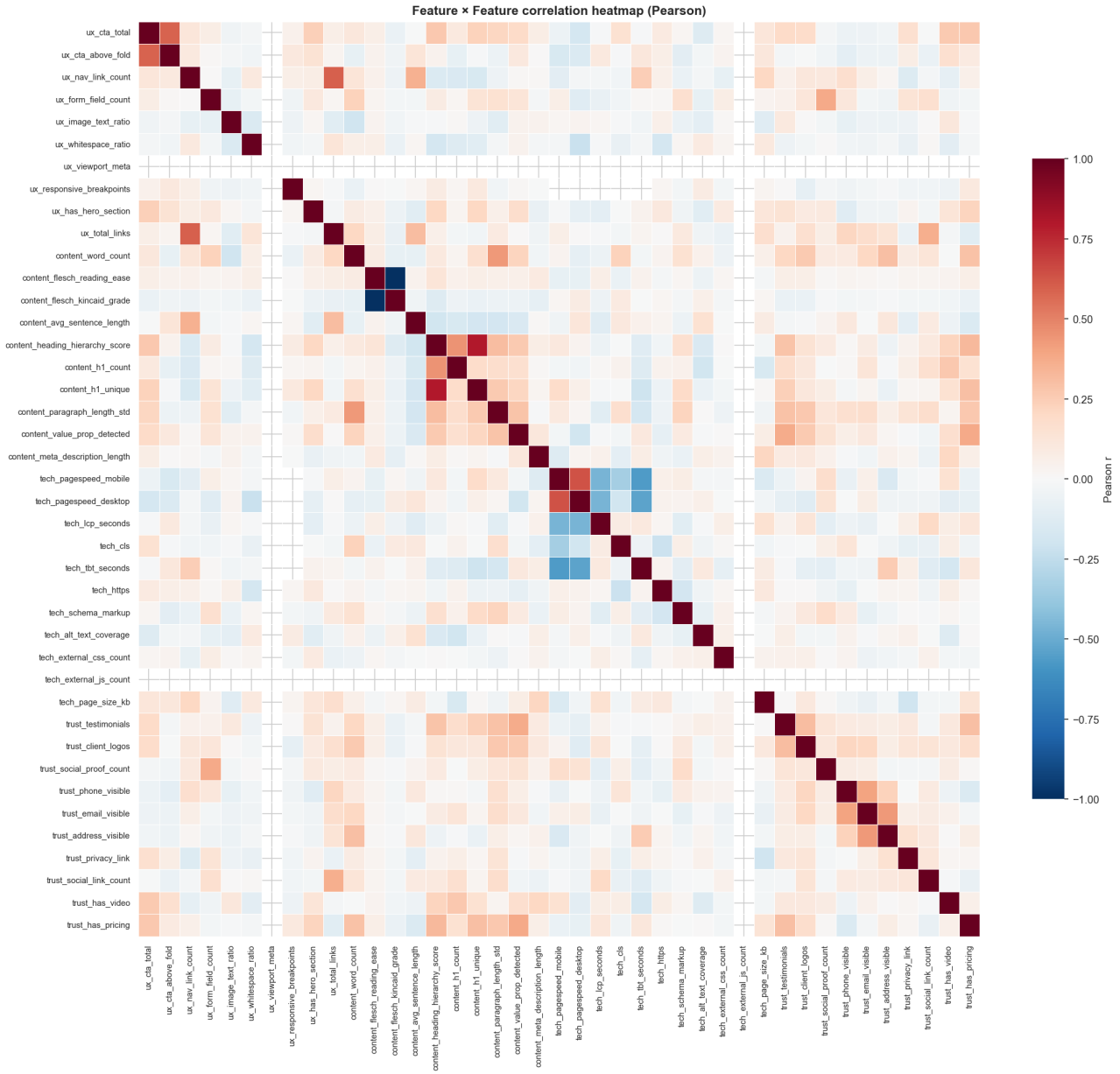

Figure 7 — Full feature × feature correlation matrix. Block structure within each dimension is visible; one perfect anti-correlation (flesch_reading_ease vs. flesch_kincaid_grade).

08 feature target correlation

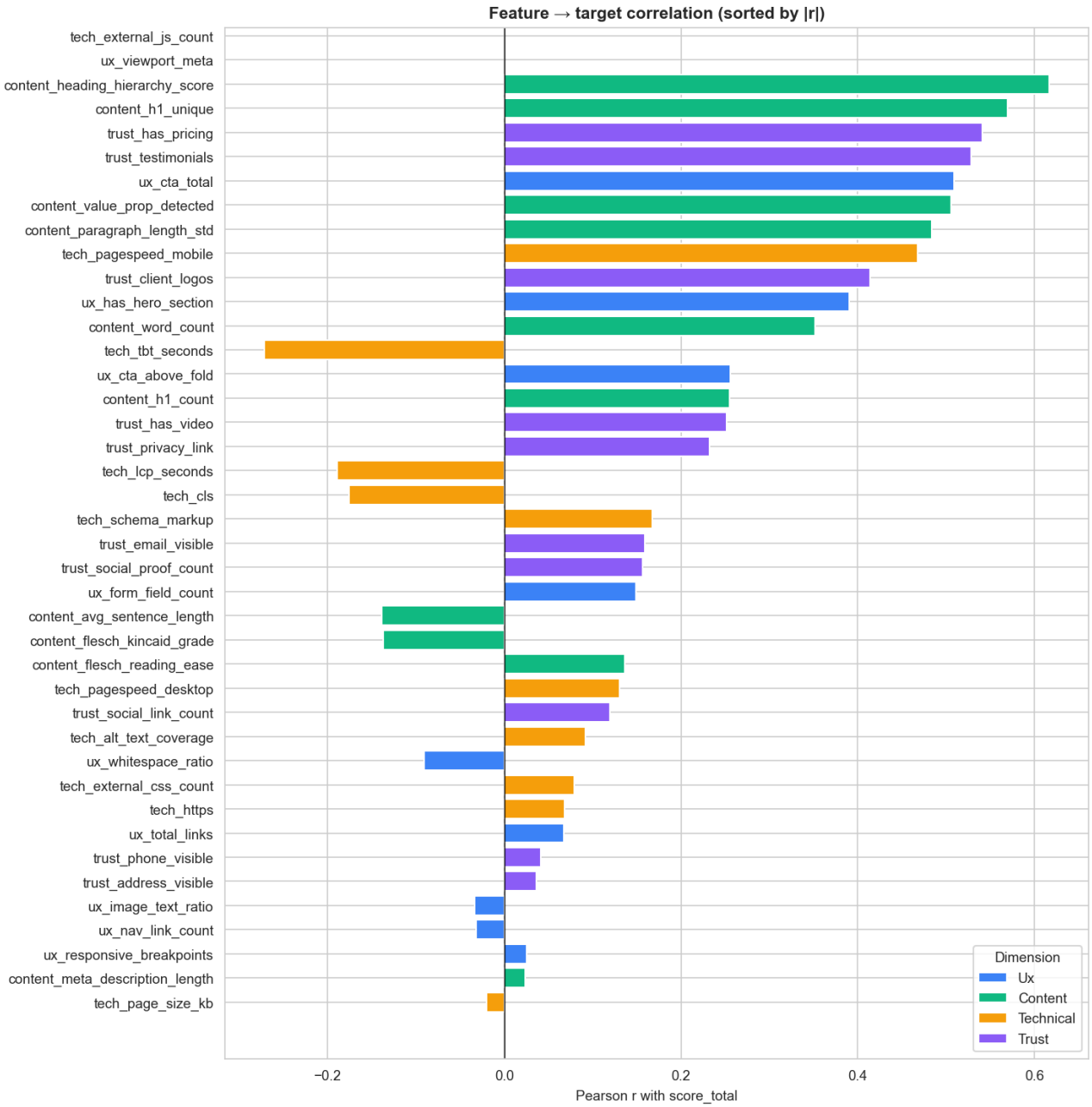

Figure 8 — Each feature's Pearson r with score_total, sorted by magnitude, colored by dimension. content_heading_hierarchy_score and content_h1_unique lead.

09 top features scatter

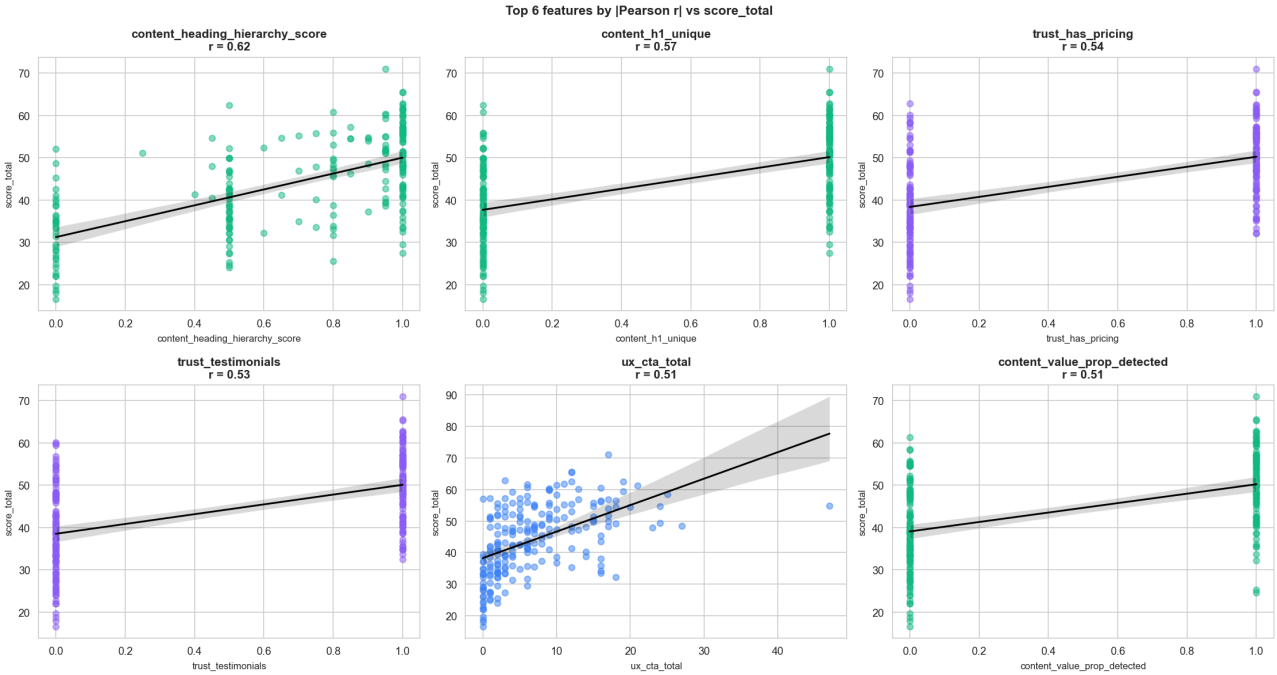

Figure 9 — Scatter plots of the top-6 most-correlated features against score_total with regression lines.

10 boxplots continuous

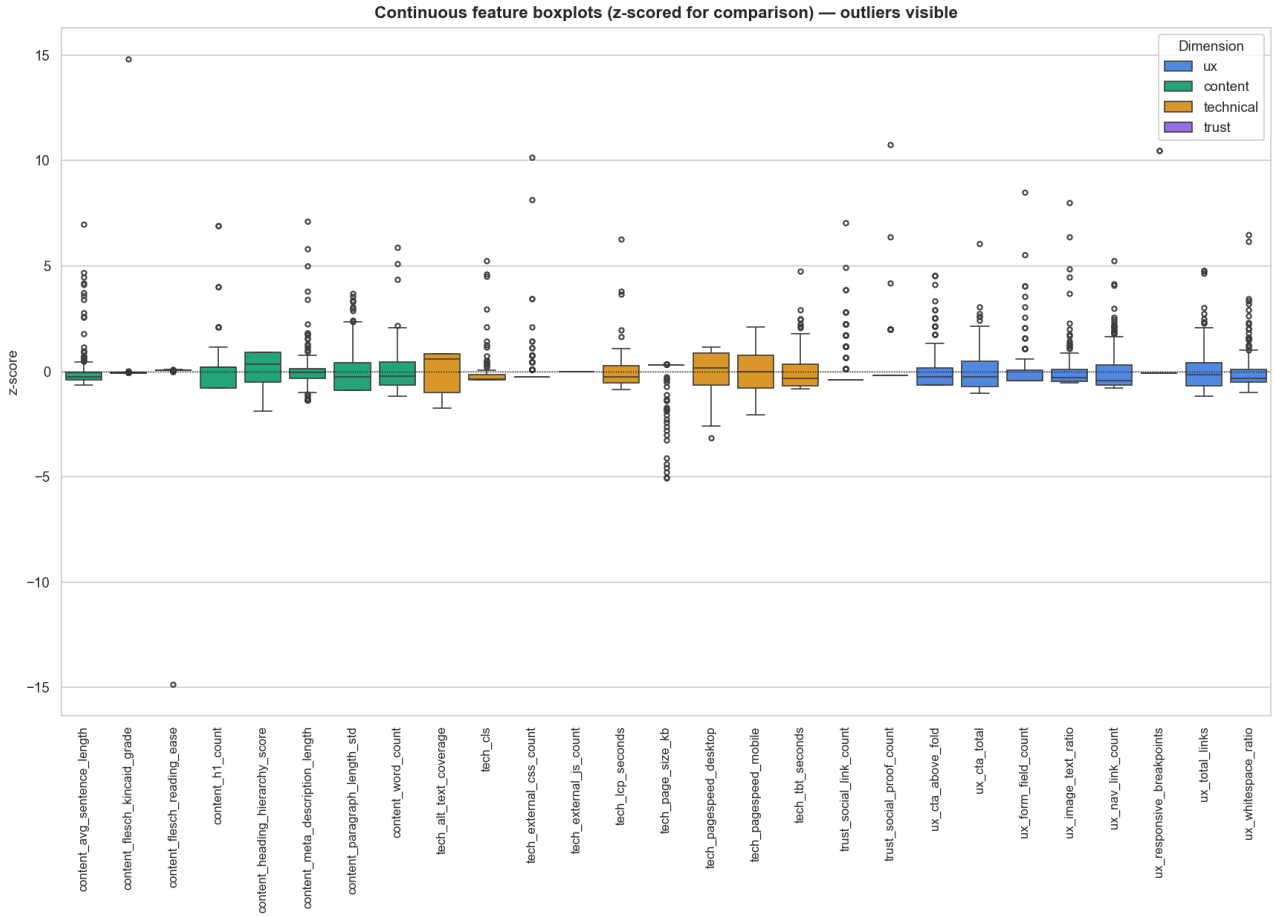

Figure 10 — Z-scored boxplots of all continuous features. Outliers are visible as points beyond whiskers; high-skew features (e.g., page_size_kb, word_count) dominate.

11 subscore pairplot

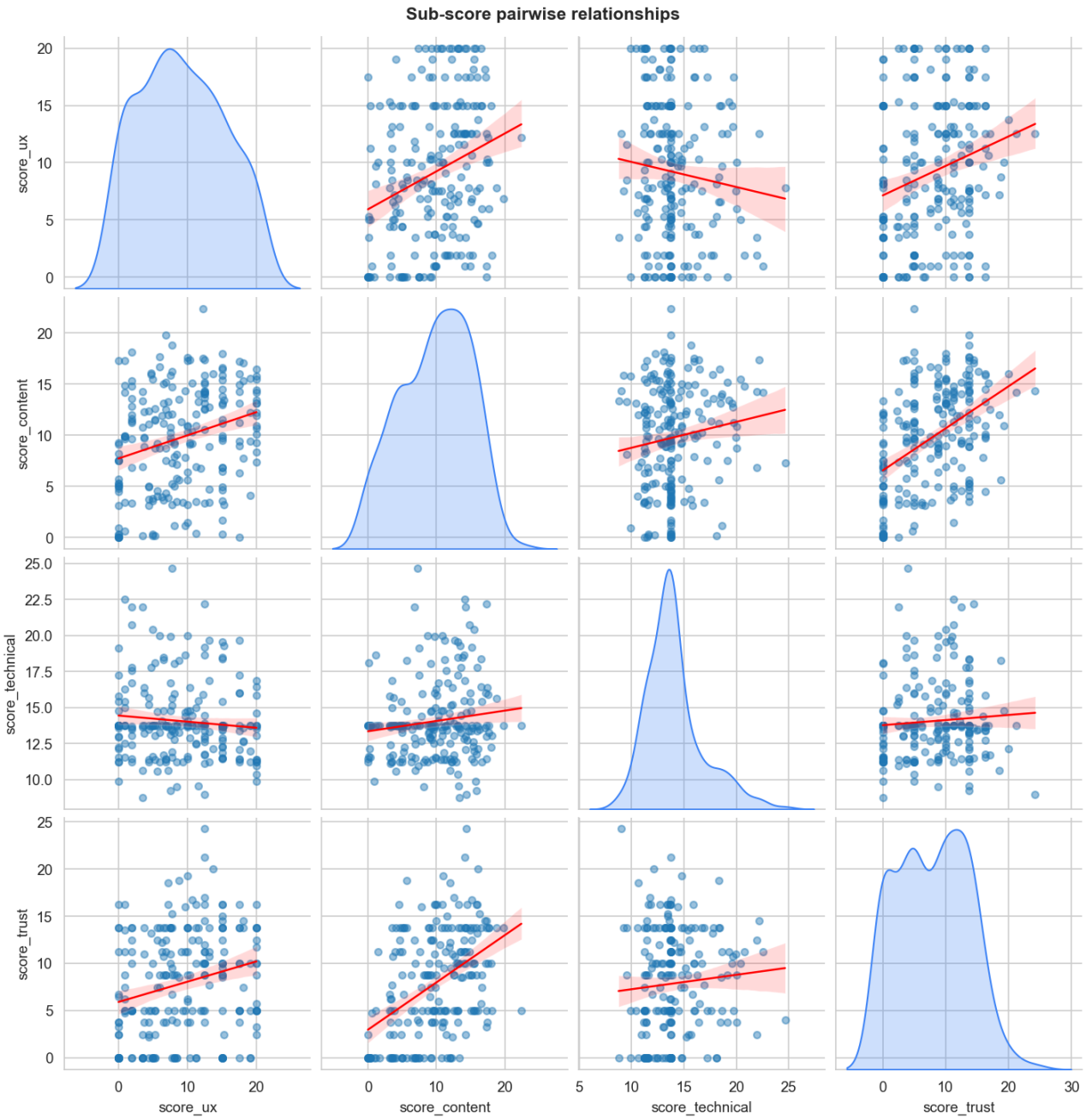

Figure 11 — Pairwise relationships among the four sub-scores. Content↔Trust is the strongest (r=0.45); Technical is largely independent.

12 skewness bars

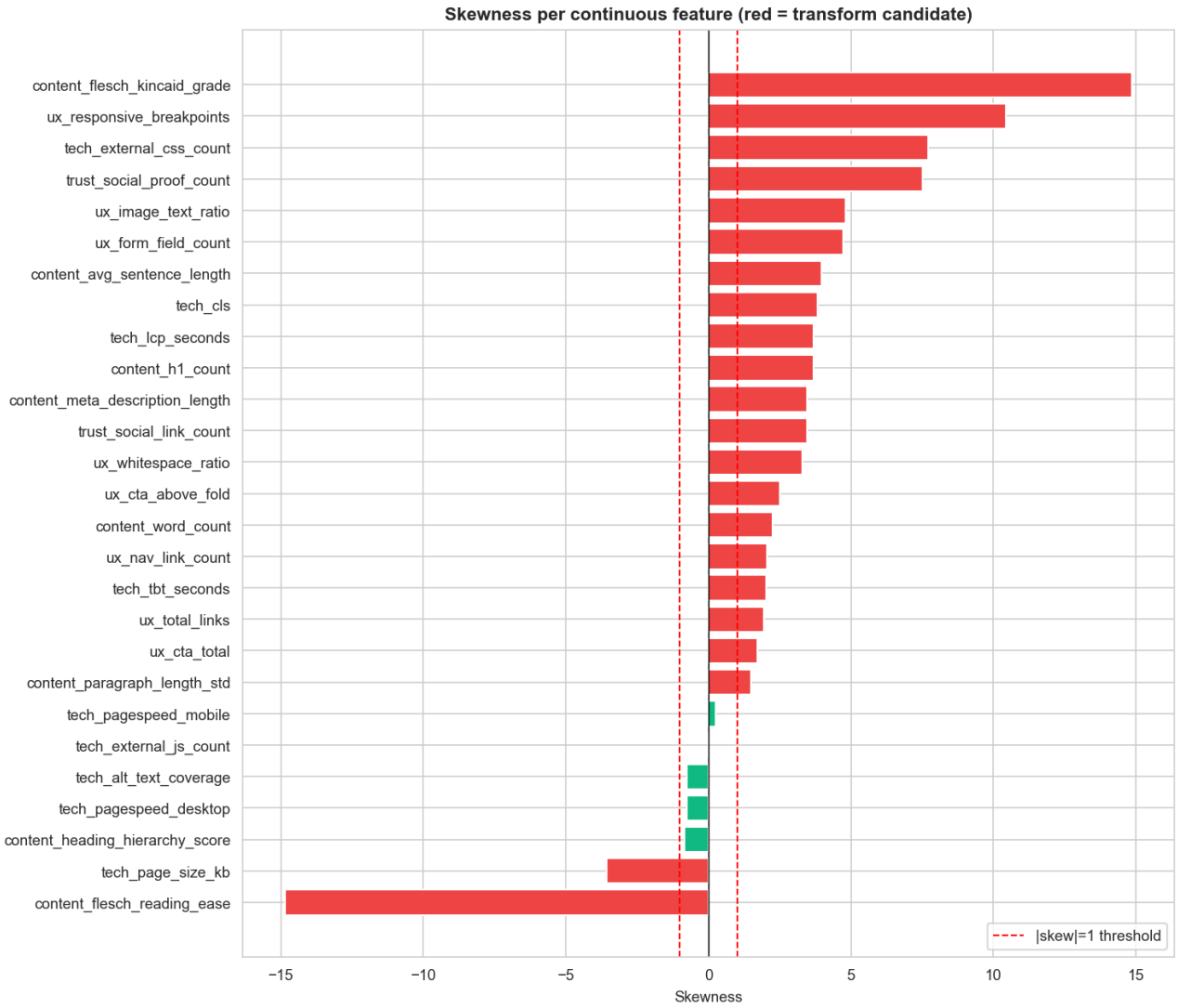

Figure 12 — Skewness per continuous feature. Red bars (|skew| > 1) are log-transform candidates; 22 features qualify.

13 pca scatter

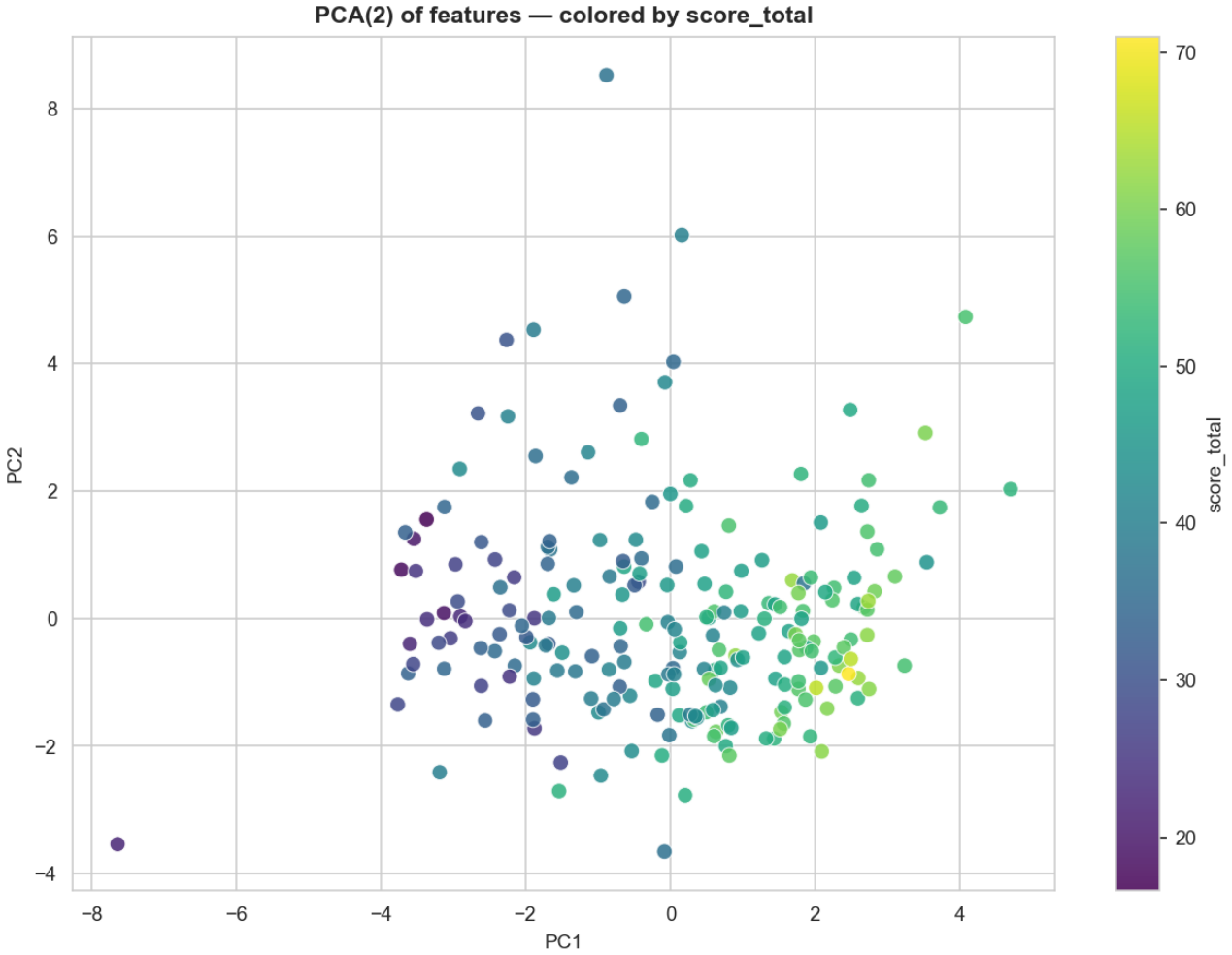

Figure 13 — 2D PCA projection of features, colored by score_total. Gradient along PC1 confirms the feature space carries a meaningful score signal.

14 score total by boolean

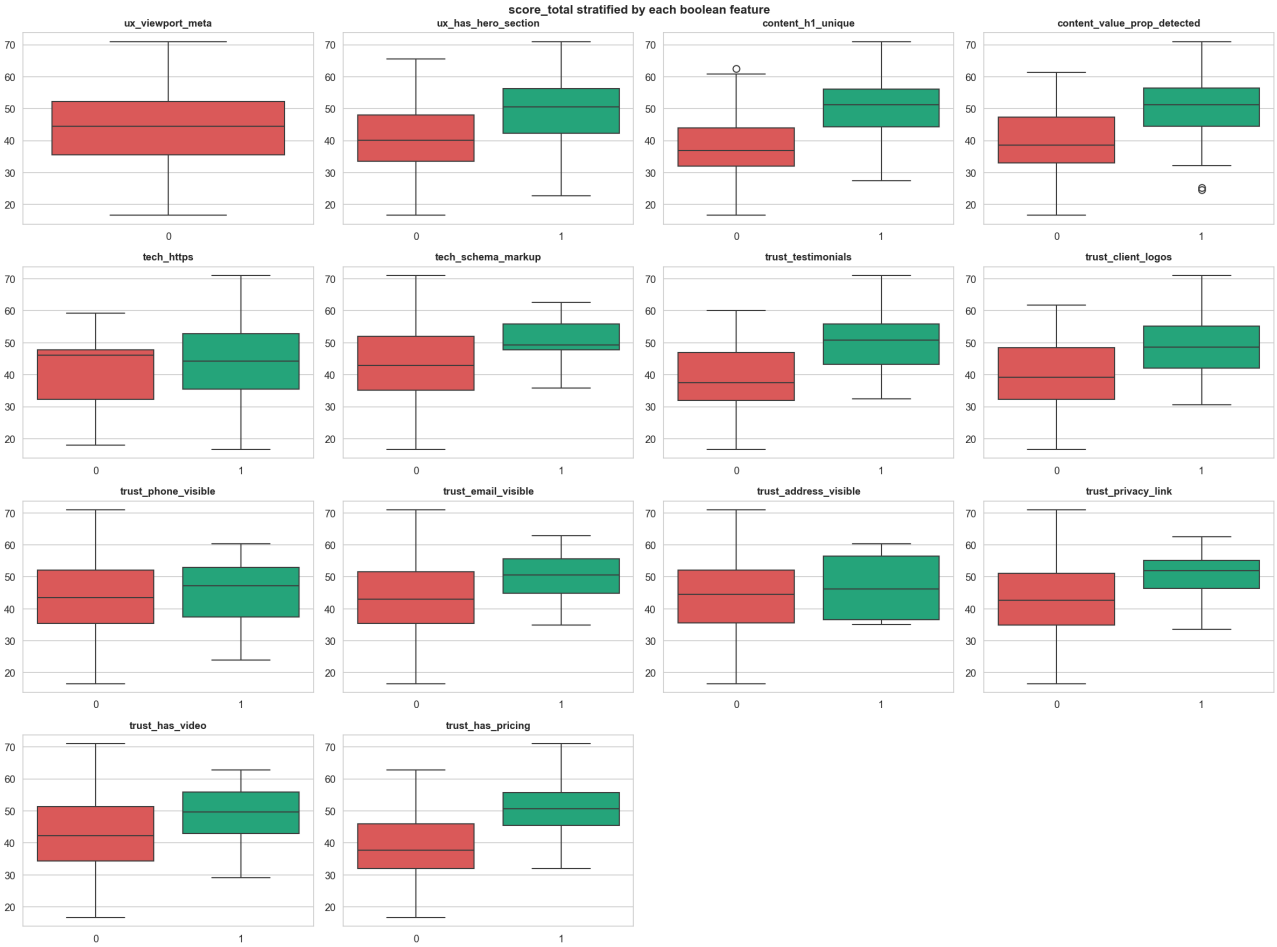

Figure 14 — score_total stratified by each boolean feature (True vs. False). Boolean features where the green box sits clearly above the red one contribute strong positive signal.